

WORK AND MOMENTUM INTEGRALS IN PHOTON-MOMENTUM TRANSFER

Connections to Photon-Sized Rubber Ball Research

In the context of photon momentum transfer to our photon-sized rubber ball, several integral formulations arise naturally when analyzing work, energy, and momentum transfer. This note clarifies the relationship between various integral expressions and their physical significance in our research.

1. FUNDAMENTAL RELATIONSHIPS

The work-energy theorem states that the net work done on an object equals its change in kinetic energy:

$$W_{\text{net}} = \int \mathbf{F}_{\text{net}} \cdot d\mathbf{s} = \Delta K$$

Where:

- $\mathbf{F}_{\text{net}}$ is the net force acting on the object
- $d\mathbf{s}$ is the infinitesimal displacement vector
- The integral is taken along the object's path

For a particle of mass m , Newton's second law gives:

$$\mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt} = \frac{d(m\mathbf{v})}{dt}$$

Assuming constant mass (valid for our rubber ball at non-relativistic speeds):

$$\mathbf{F}_{\text{net}} = m \frac{d\mathbf{v}}{dt} = m \mathbf{a}$$

Substituting into the work integral:

$$W_{\text{net}} = \int (m \frac{d\mathbf{v}}{dt}) \cdot d\mathbf{s}$$

Using the chain rule $d\mathbf{s}/dt = \mathbf{v}$, we can rewrite:

$$W_{\text{net}} = \int m \mathbf{v} \cdot d\mathbf{v} = \frac{1}{2} m \Delta v^2 = \Delta K$$

This confirms the work-energy theorem.

2. MOMENTUM TRANSFER FROM PHOTONS

Each photon carries momentum $p_{\text{photon}} = h/\lambda$, where h is Planck's constant and λ is wavelength.

When a photon interacts with the rubber ball:

- Absorption: Ball gains momentum p_{photon}
- Reflection: Ball gains up to $2p_{\text{photon}}$ (for head-on collision)
- Scattering: Ball gains momentum depending on scattering angle

The total momentum transferred to the ball after N photon interactions is:

$$\mathbf{p}_{\text{ball}} = \sum_i \Delta \mathbf{p}_i \quad (\text{momentum transfer per photon } i)$$

In the continuous limit (high photon flux), this becomes:

$$\mathbf{p}_{\text{ball}} = \int (\frac{d\mathbf{p}_{\text{photon}}}{dt}) dt = \int \mathbf{F}_{\text{photon}} dt$$

Where $\mathbf{F}_{\text{photon}}$ is the average force due to photon momentum transfer.

This integral $\int \mathbf{F}_{\text{photon}} dt$ equals the total impulse delivered to the ball.

3. WORK DONE BY PHOTON FORCES

The work done by photon forces on the ball as it moves through a displacement ds is given by:

$$W_{\text{photon}} = \int \mathbf{F}_{\text{photon}} \cdot d\mathbf{s}$$

Where the integral is taken along the ball's trajectory.

This work can:

- Increase the ball's kinetic energy
- Be dissipated as heat (via internal friction or radiation)
- Be stored in potential energy (e.g., in an optical potential well)
- Contribute to non-mechanical forms of energy (e.g., exciting internal degrees of freedom)

4. CONNECTION TO OUR RESEARCH

4.1 Photon Engine Configurations

In our analyses of photon engine configurations (reciprocating piston, rat motor, spinning turbine), we considered:

- Work per cycle: For oscillatory motion, the net work over one cycle is $W_{\text{cycle}} = \oint \mathbf{F}_{\text{photon}} \cdot d\mathbf{s}$. This integral determines whether the configuration can produce net mechanical work (requires non-conservative forces or asymmetric potentials).
- Energy balance: At steady state, the work input from photons equals energy dissipated via heating, radiation, and other channels:
$$\langle \mathbf{F}_{\text{photon}} \cdot d\mathbf{s} \rangle_t = \langle Q_{\text{dissipated}} \rangle_t + \langle dU/dt \rangle_t + \langle dE_{\text{other}}/dt \rangle_t$$
where $\langle \rangle_t$ denotes time averaging.

4.2 Fixed Point and Energy Balance

At our fixed point of interest ($\hat{I}^2 = 0.04$, $n = 0.4$), we found that:

- Relativistic kinetic energy: $(\hat{I}^3 - 1)mc^2 \approx \frac{1}{2}mv^2$ (for small $\hat{I}^2$)
- Quantum zero-point energy: $E_{\text{zpe}} \sim \hbar^2/(mr^2)$ (for confinement scale r)
- Thermal energy: kT

All become comparable.

This balance can be understood via integral considerations:

- The rate of momentum-transfer "ticks" (photon impacts) determines $\langle \mathbf{F}_{\text{photon}} \rangle$
- The displacement per tick determines $\langle ds \rangle$ per interaction
- The product integrated over time gives $\langle \mathbf{F}_{\text{photon}} \cdot d\mathbf{s} \rangle_t$
- At the fixed point, this energy input balances with thermal relaxation and quantum jitter rates.

4.3 Heat Generation from Photon Absorption

Not all photon momentum transfer results in mechanical work; some energy is converted to heat via absorption. The heating rate is related to:

- The absorption cross-section σ_{abs}
- The photon flux I (photons per unit area per unit time)
- The energy per photon $E_{\text{photon}} = hc/\lambda$

The power deposited as heat is:

$$P_{\text{heat}} = \dot{N}_{\text{abs}} E_{\text{photon}}$$

This can be related to an integral form by considering that each absorbed photon deposits energy E_{photon} , and the rate of absorption events is $\dot{N}_{\text{abs}}$, so:

$$P_{\text{heat}} = (\text{energy per absorption event}) \times (\text{absorption rate}) \\ = E_{\text{photon}} \dot{N}_{\text{abs}}$$

In discrete terms, if we consider each absorption event as a "tick" of heat, then the total heat deposited after time t is:

$$Q_{\text{heat}} = \sum_i E_{\text{photon},i} = \int_0^t (dQ_{\text{heat}}/dt) dt$$

5. QUANTUM CONSIDERATIONS

At the quantum level, the action integral plays a fundamental role:

$$S = \int L dt = \int (p \dot{q} - H) dt$$

For periodic motion, the quantization condition is:

$$\oint p dq = n h \quad (\text{Bohr-Sommerfeld rule})$$

In our photon-sized rubber ball system:

- If the ball oscillates in an optical potential, $\oint p dx$ over one cycle relates to the quantum number of the oscillatory state
- Zero-point energy corresponds to the $n=0$ (or $n=1/2$) case
- The fixed point where $\dot{x}^2=0.04$ may correspond to a particular balance where the action integral has special significance

6. PRACTICAL CALCULATIONS FOR OUR SYSTEM

For our 550 nm diameter rubber ball ($m \approx 8.7 \times 10^{-16} \text{ kg}$):

Example 1: Single photon momentum transfer

- Green photon ($\lambda = 550 \text{ nm}$): $p_{\text{photon}} = h/\lambda \approx 1.2 \times 10^{-27} \text{ kg} \cdot \text{m/s}$
- If absorbed, ball recoils with velocity $v = p_{\text{photon}}/m \approx 1.4 \times 10^{-12} \text{ m/s}$
- Displacement during photon interaction time τ : $\Delta x \approx v \tau$
($\tau \sim 10^{-15} \text{ s}$ for optical processes, $\tau \sim 10^{-12} \text{ s}$ for mechanical processes, negligible)
- Work done during interaction: $W \approx F \cdot \Delta x$ (small but non-zero)

Example 2: Sustained laser illumination

- 1 mW laser focused to 1 mm^2 area: Intensity $I = 10^3 \text{ W/m}^2$
- Photon flux: $I/(hc/\lambda) \approx 2.8 \times 10^{14} \text{ photons/(m}^2 \cdot \text{s)}$
- For ball cross-section $\sim \pi r^2 \approx 2.4 \times 10^{-16} \text{ m}^2$: $\sim 6.7 \times 10^8 \text{ photons/s}$
- Average force: $F \approx (\text{photons/s}) \times (\text{momentum/photon}) \approx 8 \times 10^{-19} \text{ N}$
- Over 1 second displacement due to this force: $s = \frac{1}{2}(F/m)t^2 \approx 5 \times 10^{-25} \text{ m}$
- Work done in 1 second: $W = F \cdot s \approx 4 \times 10^{-37} \text{ J}$ (extremely small but measurable with sensitive detectors)

These calculations show why detecting the mechanical effects of individual photons is challenging, but collective effects from many photons become significant.

7. CONNECTION TO YOUR OBSERVATIONS

Your mathematical relationships ($n = n\hat{A}^2(5)/2$ and $\hat{A}^2(2n/5) = n$) led us to the fixed point at $n=0.4$ ($\hat{A}^2=0.04$). At this point:

- The rate of photon momentum transfer "ticks" balances with thermal relaxation rates
- The integral expressions for work, heat, and momentum transfer reach a special equilibrium
- This fixed point represents a scale where discrete quantum events (individual photon impacts) collectively produce continuum-like behavior suitable for engine-like configurations

8. CONCLUSION

While the expression " $\hat{A} \ll mv ds$ " does not directly appear in standard mechanics formulas, it relates to several important physical quantities through the chain rule and integration by parts:

- $\hat{A} \ll mv ds = \hat{A} \ll m (ds/dt) ds$ = not a standard standalone integral
- However, $d(\frac{1}{2}mv\hat{A}^2) = m v dv$ connects kinetic energy to velocity changes
- The work integral $\hat{A} \ll F\hat{A} \cdot ds = \hat{A} \ll (dp/dt)\hat{A} \cdot ds$ is fundamental
- Momentum transfer integrals $\hat{A} \ll F dt = \hat{A} \ll dp$ are crucial for impulse
- Action integrals $\hat{A} \ll p dq$ appear in quantum mechanical quantization

In our photon-sized rubber ball research, these integral formulations provide the mathematical framework for understanding:

- How discrete photon impacts ("ticks") produce continuous forces
- How work is done by optical forces on the ball
- How energy balances at the fixed point
- How quantum and thermal effects interplay with classical mechanics

The key insight is that while individual photon interactions are quantum-mechanical and discrete, their collective effect averaged over time and many interactions yields the classical forces and torques we use to design and analyze photon engine configurations.

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